

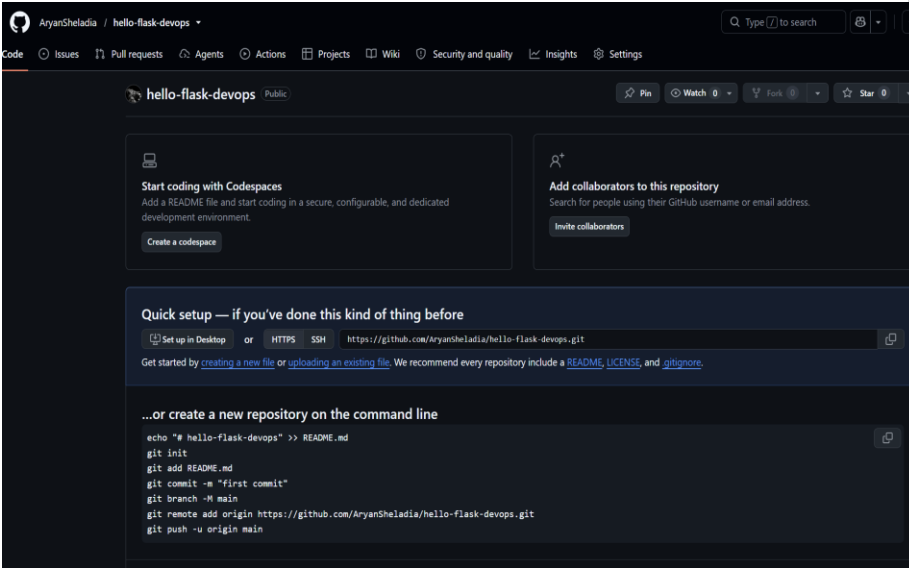


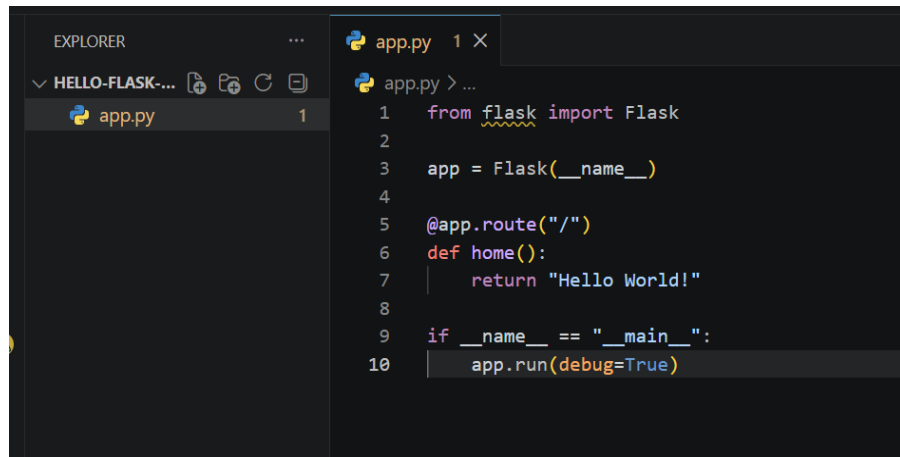
SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

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Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 1	
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Batch	2023 - 2027
Class	CSE – B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Git Setup & Basic Workflow
Implementation & Theory	 Empty GitHub Repository Created a new empty repository named hello-flask-devops on GitHub without initializing it with a README, .gitignore, or license. This allows the local project to be connected and managed using Git from scratch.



```
EXPLORER
HELLO-FLASK-...
app.py 1

app.py 1 X
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def home():
7     return "Hello World!"
8
9 if __name__ == "__main__":
10     app.run(debug=True)
```

Created Flask Application

Created a local project directory named **hello-flask-devops** and added a simple Flask application (app.py) that displays **"Hello World!"**. This serves as the initial project that will be tracked using Git.

```
PS E:\SEM 7\DevOpsLab\hello-flask-devops> git init
Initialized empty Git repository in E:/SEM 7/DevOpsLab/hello-flask-devops/.git/
PS E:\SEM 7\DevOpsLab\hello-flask-devops>
```

Initialized Git Repository

Initialized a new Git repository inside the project directory using the git init command. This creates the hidden .git folder, enabling Git to track changes made to the project files.

```
Initialized empty Git repository in E:/SEM 7/DevOpsLab/hello-flask-devops/.git/
PS E:\SEM 7\DevOpsLab\hello-flask-devops> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    app.py

nothing added to commit but untracked files present (use "git add" to track)
PS E:\SEM 7\DevOpsLab\hello-flask-devops>
```

Checked Repository Status

Used the git status command to verify the current state of the repository. Git identified **app.py** as an untracked file, indicating that it has not yet been added to the staging area for version control.

```

nothing added to commit but untracked files present (use
• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git add .
• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   app.py

```

Staged Project Files

Used the `git add .` command to add all project files to the Git staging area. Running `git status` confirmed that **app.py** is staged and ready to be included in the first commit.

```

• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git branch -M main
• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git branch
• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git branch
• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git commit -m "Initial Flask application"
[main (root-commit) 5134b94] Initial Flask application
 1 file changed, 10 insertions(+)
 create mode 100644 app.py
• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git status
On branch main
nothing to commit, working tree clean
• PS E:\SEM 7\DevOpsLab\hello-flask-devops>

```

Created Initial Commit

Renamed the default branch to main and created the first commit containing the Flask application using the `git commit` command. The repository status confirmed a clean working tree, indicating that all tracked changes were successfully committed.

```

• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git remote add origin https://github.com/AryanSheladia/hello-flask-devops.git
• PS E:\SEM 7\DevOpsLab\hello-flask-devops> git remote -v
origin https://github.com/AryanSheladia/hello-flask-devops.git (fetch)
origin https://github.com/AryanSheladia/hello-flask-devops.git (push)

```

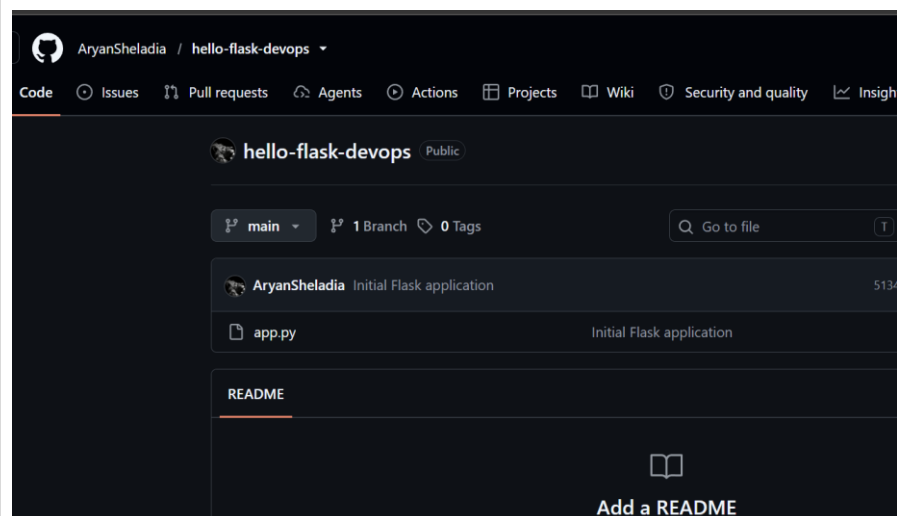
Connected Local Repository to GitHub

Configured the local Git repository to communicate with the remote GitHub repository using the `git remote add origin` command. The `git remote -v` command verified the remote repository URL for both fetch and push operations.

```
PS E:\SEM 7\DevOpsLab\hello-flask-devops> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 346 bytes | 346.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/AryanSheladia/hello-flask-devops.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS E:\SEM 7\DevOpsLab\hello-flask-devops>
```

Pushed Project to GitHub

Used the `git push -u origin main` command to upload the local Git repository to GitHub. The `-u` option established a tracking relationship between the local **main** branch and the remote **origin/main** branch for future push and pull operations.



Verified Repository on GitHub

Refreshed the GitHub repository after pushing the local changes. The presence of the **app.py** file and the latest commit confirms that the repository has been successfully synchronized with the remote GitHub repository.

